Health technology advancement: Wireless ultrasound patch

Researchers at the University of California San Diego have developed a fully wireless ultrasound patch that can continuously track critical vital signals such as heart rate and blood pressure. A flexible circuit manages ultrasound functions and wirelessly sends signals to a device. A machine learning algorithm processes these signals, tracking the artery even during motion, offering new opportunities in medical ultrasonography and exercise physiology. The researchers cross-validated a machine learning model on ten subjects from various racial groups, finding it could accurately track carotid artery pulsations, allowing measurements like blood pressure and giving early heart failure warnings. They stressed the need to test the patch on a larger population and compare it to existing devices. Besides assessing cardiovascular functions, the device was used to monitor the diaphragm and peripheral arteries, with potential for adaptable probe designs for different body areas.

The wireless ultrasound patch. The probe is placed on the carotid artery (Credit: Xu lab at UC San Diego)

Versatile skin-like sensors fit nearly anywhere

Researchers from the Munich Institute of Robotics and Machine Intelligence (MIRMI) at the Technical University of Munich have developed an automatic process for creating soft sensors. The framework uses software to design sensory systems and a 3D printer to make soft sensors by injecting conductive paste into silicone. These sensors adapt to surfaces like fingers, changing electrical resistance when squeezed or stretched, providing precise data for controlling artificial hands. This adaptability ensures accurate feedback for effective interactions. Integrating soft sensors into 3D objects enhances haptic sensing in AI, offering real-time data on forces and deformations for immediate feedback. This expands perception and enables more nuanced interactions in robotics, prosthetics, and human-machine relations. The researchers see potential for revolutionary impact, allowing wireless and customisable sensor technology across different objects and machines.

The sensor skin is very flexible and can be attached to many surfaces, including fingers, for example (Credit: Andreas Heddergott/TUM)